

Unitarity Controls Two-Copy Mixing and the Sampling Cost of Randomized Benchmarking

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Randomized benchmarking estimates a mean decay from finitely many random circuits, so its experimental cost is controlled by a two-copy transfer operator. Wallman and Flammia conjectured that this operator is contractive for every nonunitary channel. We prove the conjecture in arbitrary dimension and strengthen it to an exact Hilbert–Schmidt contraction: the two-copy core has induced norm and spectral radius equal to the channel unitarity u . Centering the channel further gives $\|T^m - f^{2m}\mathbb{I}\|_{2 \rightarrow 2} = u^m - f^{2m}$, where f is the usual benchmarking decay. This identity yields finite-length variance bounds with no Jordan condition number, rigorous sequence counts, and a direct estimator of normalized nonunitarity. We also give a noise-adapted certificate for arbitrary randomizing distributions. For an approximate 2-design with moment error ϵ_2 , two-copy mixing is guaranteed whenever $\epsilon_2 a_4 < 1 - u$, where $a_4 = \|\phi^\top \phi - u\mathbb{I}\|_2$ is the contraction anisotropy. This converts a random-circuit moment gap into a sufficient, noise-adapted randomization depth. The centered contraction also extends exactly to time-dependent Markovian noise through an ordered product formula.

Problem.— Randomized benchmarking replaces a fixed noise channel by products of randomly conjugated copies. Its long-sequence variance is governed by the peripheral spectrum of a twirled two-copy channel [1]. A channel is called *two-contractive* when the eigenvalue 1 of this twirl is algebraically simple and every other eigenvalue lies strictly inside the unit disk. Wallman and Flammia proved two-contractivity for every nonunitary unital channel and, by a special property of qubit channels, for every nonunitary qubit channel. They conjectured the same conclusion for arbitrary nonunital channels in every dimension [1]. We use this spectral definition throughout; it should not be conflated with variants of strong irreducibility that additionally require a faithful stationary state.

Let $\Lambda : M_d \rightarrow M_d$ be completely positive and trace preserving (CPTP). Choose a trace-orthonormal Hermitian operator basis $B_0 = \mathbb{I}_d/\sqrt{d}, B_1, \dots, B_D$, where $D = d^2 - 1$. The Hermitian Liouville matrix of Λ is real and has the block form

$$[\Lambda] = \begin{pmatrix} 1 & 0 \\ \alpha & \phi \end{pmatrix}, \quad \phi \in M_D(\mathbb{R}). \quad (1)$$

The vector α is the nonunital part; ϕ is the restriction to the traceless Hermitian subspace. We use the standard depolarizing parameter and unitarity

$$f = \frac{\text{Tr } \phi}{D}, \quad u = \frac{\text{Tr}(\phi^\top \phi)}{D}. \quad (2)$$

For a CPTP channel, $0 \leq u \leq 1$, and $u = 1$ if and only if Λ is a unital channel [2]. Throughout, $\|\cdot\|_2$ denotes the Euclidean norm for vectors and the Hilbert–Schmidt norm for matrices, $\|\cdot\|_\infty$ denotes the operator norm, and $\|\cdot\|_{2 \rightarrow 2}$ is the induced Hilbert–Schmidt norm.

Let $\{(p_g, U_g)\}_g$ be a finite weighted exact unitary 2-design and let $R_g \in O(D)$ be the adjoint action of U_g on the traceless Hermitian subspace. Define $\Lambda^g = \mathcal{U}_g^\dagger \circ \Lambda \circ \mathcal{U}_g$ and

$$\mathcal{M}_2 = \sum_g p_g (\Lambda^g)^{\otimes 2}. \quad (3)$$

Uniform averaging over a finite design group is the original setting of Ref. [1]; allowing weights does not change the proof.

Theorem 1 (Two-copy contractivity) *For every CPTP map $\Lambda : M_d \rightarrow M_d$ and every exact unitary 2-design, the nontrivial spectral radius of $\mathcal{M}_2$ is*

$$r_{\text{nt}}(\mathcal{M}_2) = \max\{|f|, u\} \leq \sqrt{u}. \quad (4)$$

Hence Λ is two-contractive if and only if it is nonunitary.

The conclusion resolves the conjecture in its full dimension-independent form. It is stronger than a qualitative peripheral-spectrum statement: Eq. (4) identifies the exact exponential rate that controls all nontrivial modes once f and u are known.

Proof.— The exact 2-design moment identity reproduces the Haar Schur average on the traceless adjoint representation. Thus, for every $X \in M_D(\mathbb{C})$,

$$\sum_g p_g R_g^\top X R_g = \frac{\text{Tr } X}{D} \mathbb{I}_D. \quad (5)$$

Taking $X = \phi$ gives $\sum_g p_g R_g^\top \phi R_g = f \mathbb{I}_D$. In the ordered block basis $\mathbb{R} \oplus (\mathbb{R} \otimes V) \oplus (V \otimes \mathbb{R}) \oplus (V \otimes V)$, with $V = \mathbb{R}^D$, Eq. (1) makes $\mathcal{M}_2$ block lower triangular. Its diagonal blocks are

$$1, \quad f \mathbb{I}_D, \quad f \mathbb{I}_D, \quad T = \sum_g p_g A_g \otimes A_g, \quad (6)$$

where $A_g = R_g^\top \phi R_g$. All dependence on the nonunital vector α occurs below the diagonal and hence cannot change the spectrum.

It remains to control T without assuming that the unital projection of Λ is itself completely positive. Identify $V \otimes V$ with $M_D(\mathbb{C})$ by vectorization. Because every A_g is real,

$$T \text{vec}(X) = \text{vec}[\mathcal{T}(X)], \quad \mathcal{T}(X) = \sum_g p_g A_g X A_g^\top \quad (7)$$

defines a completely positive map on the auxiliary matrix algebra. A second Schur average yields

$$\mathcal{T}(\mathbb{I}_D) = \sum_g p_g R_g^\top (\phi \phi^\top) R_g = \frac{\text{Tr}(\phi \phi^\top)}{D} \mathbb{I}_D = u \mathbb{I}_D. \quad (8)$$

The same calculation with $\phi^\top \phi$ gives $\mathcal{T}^\dagger(\mathbb{I}_D) = u\mathbb{I}_D$. If $u = 0$, then $\phi = 0$ and $T = 0$. If $u > 0$, the map $\mathcal{T}/u$ is bistochastic and completely positive. Kadison's inequality and trace preservation imply

$$\|\mathcal{T}(X)\|_2^2 \leq u \operatorname{Tr}[\mathcal{T}(X^\dagger X)] = u^2 \|X\|_2^2. \quad (9)$$

The identity operator attains the bound, and hence

$$\|T\|_{2 \rightarrow 2} = \rho(T) = u, \quad \|T^m\|_{2 \rightarrow 2} = u^m. \quad (10)$$

In particular every eigenvalue λ of T satisfies $|\lambda| \leq u$; Eq. (8) shows that u itself is an eigenvalue. The induced-norm identity is stronger than the spectral claim and will remove every unknown Jordan prefactor from the finite-length bounds below.

Finally, Hilbert–Schmidt Cauchy–Schwarz gives

$$|f| = \frac{|\operatorname{Tr} \phi|}{D} \leq \sqrt{\frac{\operatorname{Tr}(\phi^\top \phi)}{D}} = \sqrt{u}. \quad (11)$$

The spectrum of a block triangular matrix is the union of the spectra of its diagonal blocks. Combining Eqs. (6), (10), and (11) proves Eq. (4). A nonunitary CPTP map has $u < 1$, so the scalar top-left eigenvalue is algebraically simple and is the unique eigenvalue of unit modulus. Conversely, a unitary channel has $u = 1$, and the eigenvector $\operatorname{vec}(\mathbb{I}_D)$ supplies another unit-modulus eigenvalue. This completes the proof.

The auxiliary map also controls the difference between the two-copy and independent one-copy decays. This difference, rather than T alone, enters the randomized-benchmarking variance.

Lemma 1 (Centered contraction) *For every integer $m \geq 1$,*

$$\|T^m - f^{2m}\mathbb{I}_{D^2}\|_{2 \rightarrow 2} = u^m - f^{2m}. \quad (12)$$

Proof.— Set $C_g = A_g - f\mathbb{I}_D$. Since $\sum_g p_g A_g = f\mathbb{I}_D$,

$$T - f^2\mathbb{I}_{D^2} = \sum_g p_g C_g \otimes C_g \quad (13)$$

is the Liouville matrix of the completely positive map $\mathcal{S}(X) = \sum_g p_g C_g X C_g^\top$. Equations (5) and (8) give $\mathcal{S}(\mathbb{I}_D) = \mathcal{S}^\dagger(\mathbb{I}_D) = (u - f^2)\mathbb{I}_D$. Because $T = f^2\mathbb{I}_{D^2} + \mathcal{S}$, the binomial expansion shows that $T^m - f^{2m}\mathbb{I}_{D^2}$ is completely positive. It and its adjoint map the identity to $(u^m - f^{2m})\mathbb{I}_D$. Repeating Eq. (9) proves the upper bound in Eq. (12); the identity attains it. $\square$

Corollary 1 (Nonstationary noise) *Suppose a predetermined channel at step j may vary, while the toggling-frame conjugators are sampled independently from an exact unitary 2-design. If T_j , f_j , and u_j denote its two-copy core, depolarizing parameter, and unitarity, respectively, then*

$$\left\| T_m \cdots T_1 - \left(\prod_{j=1}^m f_j^2 \right) \mathbb{I}_{D^2} \right\|_{2 \rightarrow 2} = \prod_{j=1}^m u_j - \prod_{j=1}^m f_j^2. \quad (14)$$

No commutativity or stationarity assumption is required.

Indeed, each $T_j - f_j^2\mathbb{I}_{D^2}$ represents a completely positive map. Expanding the ordered product shows that the difference in Eq. (14) is completely positive; it and its adjoint map the identity to the scalar on the right-hand side. The bistochastic contraction argument then gives equality. Thus the core part of the finite-length fluctuation bound remains directly controlled under deterministic drift.

Finite-length variance.— We now specialize to standard randomized benchmarking with uniform sampling from a finite unitary-2-design group and gate-independent Markovian noise. Absorb the final inversion noise into the measurement and write

$$|\rho\rangle = \begin{pmatrix} \rho_0 \\ \mathbf{r} \end{pmatrix}, \quad (E| = (E_0 \quad e^\top), \quad \rho_0 = d^{-1/2}. \quad (15)$$

Let $a = \|\alpha\|_2$, $s = \|\mathbf{r}\|_2$, $e = \|e\|_2$, $q = |f|$, and define

$$H_m(x, y) = \sum_{t=0}^{m-1} x^{m-1-t} y^t = \begin{cases} (x^m - y^m)/(x - y), & x \neq y, \\ mx^{m-1}, & x = y. \end{cases} \quad (16)$$

Theorem 2 (Variance and convergence) *For a nonunitary CPTP map, the variance σ_m^2 across random sequences obeys*

$$\sigma_m^2 \leq \sigma_\infty^2 (1 - u^m) + 2\rho_0 e^2 s a \sqrt{u - f^2} H_m(q, u) + e^2 s^2 (u^m - f^{2m}), \quad (17)$$

$$|\sigma_m^2 - \sigma_\infty^2| \leq \sigma_\infty^2 u^m + 2\rho_0 e^2 s a \sqrt{u - f^2} H_m(q, u) + e^2 s^2 (u^m - f^{2m}), \quad (18)$$

where

$$\sigma_\infty^2 = \frac{e^2 a^2}{dD(1-u)} = \frac{e^2}{d(d+1)} \eta_{\text{nu}}, \quad \eta_{\text{nu}} = \frac{a^2}{(d-1)(1-u)} \in [0, 1]. \quad (19)$$

The right side of Eq. (17) may always be replaced by its minimum with $1/4$.

The normalized quantity η_{nu} is well defined for $u < 1$. The inequality $\eta_{\text{nu}} \leq 1$ is the CPTP constraint $a^2 \leq (d-1)(1-u)$ [2]. Equation (19) therefore turns the asymptotic variance into a calibrated measure of the fraction of the channel's purity-loss budget carried by its nonunit displacement.

To prove the theorem, use the exact block-power decomposition of Ref. [1]. The injection from the scalar block is $P_1 \alpha^{\otimes 2} = (a^2/D) \operatorname{vec}(\mathbb{I}_D)$ and $T \operatorname{vec}(\mathbb{I}_D) = u \operatorname{vec}(\mathbb{I}_D)$, which gives the first term in Eq. (17). Centering A_g as in Eq. (13) gives the two mixed-block estimates

$$\|b\|_{2 \rightarrow 2}, \|c\|_{2 \rightarrow 2} \leq a \sqrt{u - f^2}. \quad (20)$$

Indeed, Jensen's inequality and the Schur average give $\mathbb{E}_g \|C_g \mathbf{r}\|_2^2 = (u - f^2) \|\mathbf{r}\|_2^2$. Summing the lower-triangular convolution produces $H_m(q, u)$. Finally Eq. (12) bounds the traceless–traceless term. Applying Cauchy–Schwarz to the measurement vector proves both inequalities. If $\phi = f\mathbb{I}_D$, then $u = f^2$, both transient terms vanish, and the exact identity $\sigma_m^2 = \sigma_\infty^2 (1 - u^m)$ holds.

Sequence count and circuit reuse.— Assume that $|\rho\rangle$ represents a state and that $0 \leq E \leq \mathbb{I}$ is a binary survival effect. For each $n = 1, \dots, N$, sample a sequence independently and let $Y_{m,n}$ be its empirical survival probability from K conditionally independent Bernoulli shots. Set $\widehat{F}_m = N^{-1} \sum_n Y_{m,n}$ and $\overline{F}_m = \mathbb{E}Y_{m,n}$. If V_m denotes the right side of Eq. (17), capped at $1/4$, then

$$\begin{aligned} \text{Var}(Y_{m,n}) &\leq \overline{V}_{m,K} := \min\left\{\frac{1}{4}, V_m + \frac{1}{4K}\right\}, \\ \text{Var}(\widehat{F}_m) &\leq \frac{\overline{V}_{m,K}}{N}. \end{aligned} \quad (21)$$

Bennett's inequality therefore guarantees

$$\Pr[|\widehat{F}_m - \overline{F}_m| \geq \varepsilon] \leq 2 \exp\left[-N \overline{V}_{m,K} b\left(\frac{\varepsilon}{\overline{V}_{m,K}}\right)\right], \quad (22)$$

where $b(x) = (1+x)\ln(1+x) - x$. Thus it suffices to choose

$$N \geq \frac{\ln(2/\delta)}{\overline{V}_{m,K} b(\varepsilon/\overline{V}_{m,K})}. \quad (23)$$

Unlike a boundedness-only confidence interval, this prescription uses the experimentally accessible f and u and a calibrated nonunitarity.

The same bound makes the circuit-reuse tradeoff explicit. If compiling a new random circuit costs c_{circ} and each repeated shot costs c_{shot} , the uncapped fixed-cost mean-square relaxation is minimized at

$$K_{\text{opt}} = \frac{1}{2} \sqrt{\frac{c_{\text{circ}}}{c_{\text{shot}} V_m}}. \quad (24)$$

The implemented K is an integer at least one; one evaluates the adjacent integers, and separately checks the regime in which the $1/4$ cap is active. General circuit-reuse optimization was developed in Ref. [3]; Eq. (17) supplies its channel-dependent variance input without process tomography.

Figure 1 illustrates the resulting scale for the qutrit reset-mixture channel $\Lambda_p(\rho) = (1-p)\rho + p|0\rangle\langle 0| \text{Tr } \rho$ at $p = 0.02$. Here $f = 1-p$, $u = f^2$, $a^2 = 2p^2$, and pure ideal preparation and measurement give $e^2 = s^2 = 2/3$. At $m = 100$, Eq. (19) gives $\sigma_m^2 = 5.51 \times 10^{-4}$. With $K = 100$ shots, $\varepsilon = 0.02$, and 99% confidence, Eq. (23) requires $N = 200$ unique sequences; the boundedness-only Hoeffding guarantee requires 6623.

Imperfect and shallow randomization.— The completely positive representation does not require an exact design. For an arbitrary probability distribution ν over unitaries, let

$$\begin{aligned} A_U &= R_U^\top \phi R_U, & B_\nu &= \mathbb{E}_\nu A_U, & T_\nu &= \mathbb{E}_\nu A_U \otimes A_U, \\ Q_L &= \mathbb{E}_\nu A_U A_U^\top, & Q_R &= \mathbb{E}_\nu A_U^\top A_U. \end{aligned} \quad (25)$$

Set $s_L = \|Q_L\|_\infty$, $s_R = \|Q_R\|_\infty$, and $s = \min\{s_L, s_R\}$. The two matrices Q_L and Q_R describe the worst-direction average squared contraction of the actual randomizer.

Theorem 3 (Noise-adapted randomizer certificate) *For every sampling distribution ν , the nontrivial spectral radius of its twirled two-copy channel satisfies*

$$r_{\text{nt}} \leq \sqrt{s}. \quad (26)$$

In particular, $s < 1$ guarantees a unique peripheral eigenvalue even when ν is not a unitary design.

Indeed, the full twirl remains block lower triangular with diagonal blocks $1, B_\nu, B_\nu, T_\nu$. Matrix Jensen inequalities give $B_\nu B_\nu^\top \leq Q_L$ and $B_\nu^\top B_\nu \leq Q_R$, so $\rho(B_\nu) \leq \sqrt{s}$. Under vectorization T_ν represents the completely positive map $X \mapsto \mathbb{E}_\nu A_U X A_U^\top$. Its value on the identity is Q_L , while the adjoint maps the identity to Q_R . The induced operator-norm bounds for the map and its adjoint imply $\rho(T_\nu) \leq s$. Hence the two diagonal estimates give $r_{\text{nt}} \leq \max\{\sqrt{s}, s\} = \sqrt{s}$ when $s \leq 1$. When $s \geq 1$, the complete twirl is CPTP and therefore has spectral radius one, which again gives $r_{\text{nt}} \leq 1 \leq \sqrt{s}$. This proves Eq. (26). Operationally,

$$\mathbf{x}^\top Q_R \mathbf{x} = \mathbb{E}_\nu \|A_U \mathbf{x}\|_2^2, \quad \mathbf{y}^\top Q_L \mathbf{y} = \mathbb{E}_\nu \|A_U^\top \mathbf{y}\|_2^2, \quad (27)$$

so s can be bounded from a calibrated directional-contraction experiment or from a sampler's moment gap.

For the latter application, define the adjoint second-moment error

$$\epsilon_2 = \|\mathcal{G}_\nu - \mathcal{G}_{\text{Haar}}\|_{2 \rightarrow 2}, \quad \mathcal{G}_\nu(X) = \mathbb{E}_\nu R_U^\top X R_U, \quad (28)$$

and define the centered anisotropies

$$\begin{aligned} a_2 &= \|\phi - f \mathbb{I}_D\|_2 = \sqrt{D(u - f^2)}, \\ a_4 &= \|\phi^\top \phi - u \mathbb{I}_D\|_2 = \sqrt{\text{Tr}[(\phi^\top \phi)^2] - Du^2}. \end{aligned} \quad (29)$$

Compression of the unitary second-moment operator gives

$$s_L, s_R \leq u + \epsilon_2 a_4. \quad (30)$$

Consequently

$$\epsilon_2 a_4 < 1 - u \quad (31)$$

is a sufficient, noise-dependent design threshold, with $r_{\text{nt}} \leq \sqrt{u + \epsilon_2 a_4}$. Retaining the centered one-copy bound gives the sharper estimate

$$r_{\text{nt}} \leq R(\epsilon_2) := \max\left\{u + \epsilon_2 a_4, \min(|f| + \epsilon_2 a_2, \sqrt{u + \epsilon_2 a_4})\right\}. \quad (32)$$

For a nonunitary reference channel, $u < 1$. If an L -layer reversible random-circuit sampler has $\epsilon_2(L) \leq C(1-\gamma)^L$, with $C > 0$ and $0 < \gamma < 1$, then for $a_4 > 0$ the minimum sufficient integer depth is

$$L_{\text{min}} = \max\left\{0, \left\lceil \frac{\ln[C a_4 / (1-u)]}{-\ln(1-\gamma)} \right\rceil + 1 \right\}. \quad (33)$$

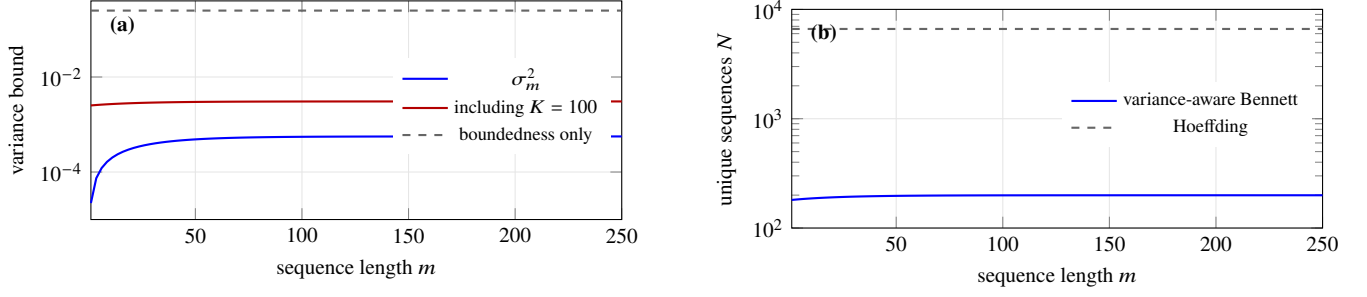


FIG. 1. Finite-sample consequence for a qutrit reset-mixture channel. (a) The exact sequence variance and the variance including $K = 100$ shots per sequence remain far below the generic $1/4$ bound. (b) The corresponding rigorous number of unique sequences for additive precision 0.02 at 99% confidence. The variance-aware guarantee reduces the boundedness-only count by more than one order of magnitude.

If $a_4 = 0$, every depth satisfies Eq. (31). This converts a random-circuit moment gap into a noise-adapted randomization depth, rather than demanding a fixed design accuracy for every device [4, 5].

Finally, consider a factorized protocol that independently samples an exact-design label g at each step and whose one- and two-copy toggling-frame moments factorize. For gate-correlated traceless blocks ϕ_g , set $A_g = R_g^\top \phi_g R_g$ and use the reference $A_g^{(0)} = R_g^\top \phi_g R_g$. The same method gives a sharp robustness radius. If $\sup_g \|\phi_g - \phi\|_\infty \leq \zeta$, then

$$r_{\text{nt}} \leq \max\{|f| + \zeta, u + 2\|\phi\|_\infty \zeta + \zeta^2\}. \quad (34)$$

The gap scaling cannot be improved in general: for a qubit, take $\phi = c\mathbb{I}_3$ and $B = \text{diag}(c, c, 1)$, with $0 < c < 1$, and set $\phi_g = R_g B R_g^\top$. Every gate error is a nonunitary dephasing channel and its average unitarity is $(1 + 2c^2)/3 < 1$, but $A_g = B$ has an exact unit eigenvalue. Here $\zeta = 1 - c$, precisely the reference one-copy gap. Equation (34) concerns protocols whose one-step toggling-frame averages factorize; ordinary gate-dependent RB generally requires an enlarged convolution transfer operator [6, 7].

Relation to prior work.— The original proof for unital channels bounded the Euclidean norm of the two-copy block by using Hilbert–Schmidt contractivity of a unital channel [1]. That route does not apply directly to a nonunital channel because its traceless block can have singular values larger than one. The auxiliary map in Eq. (7) changes the relevant norm: after Schur averaging it is unital on M_D , irrespective of whether Λ is unital.

Helsen *et al.* later bounded the scalar modes of the traceless-symmetric sector of the multiqubit Clifford twirl by the unitarity [8]. Their representation-specific result is sufficient for the variance protocol studied there. The theorem above covers arbitrary dimensions and exact unitary 2-designs, requires no multiplicity-free decomposition, controls the spectrum in absolute value, and includes the antisymmetric sector. Thus it proves the original full two-contractivity statement rather than only the physically sampled symmetric sector.

Discussion.— The proof reduces a peripheral-spectrum problem on a D^2 -dimensional tensor representation to the elementary fact that a unital completely positive map is contractive. The same normalization that proves the conjecture also removes the unknown Jordan condition number from finite-length statistics. The resulting chain is operational: standard RB and unitarity RB supply f and u ; Eq. (17) turns them into a variance bound; Eq. (23) turns that variance into a sequence count; and Eq. (24) allocates repetitions between new and reused circuits. The arbitrary-sampler certificate further shows that a full design is sufficient but not necessary. What matters is whether every directional second moment contracts for the noise actually being randomized.

The present gate-dependent statement is deliberately restricted to factorized toggling-frame sampling. Extending the same dimension-independent variance and sample guarantees to the convolution transfer operator of ordinary gate-dependent RB is the remaining application-level problem.

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